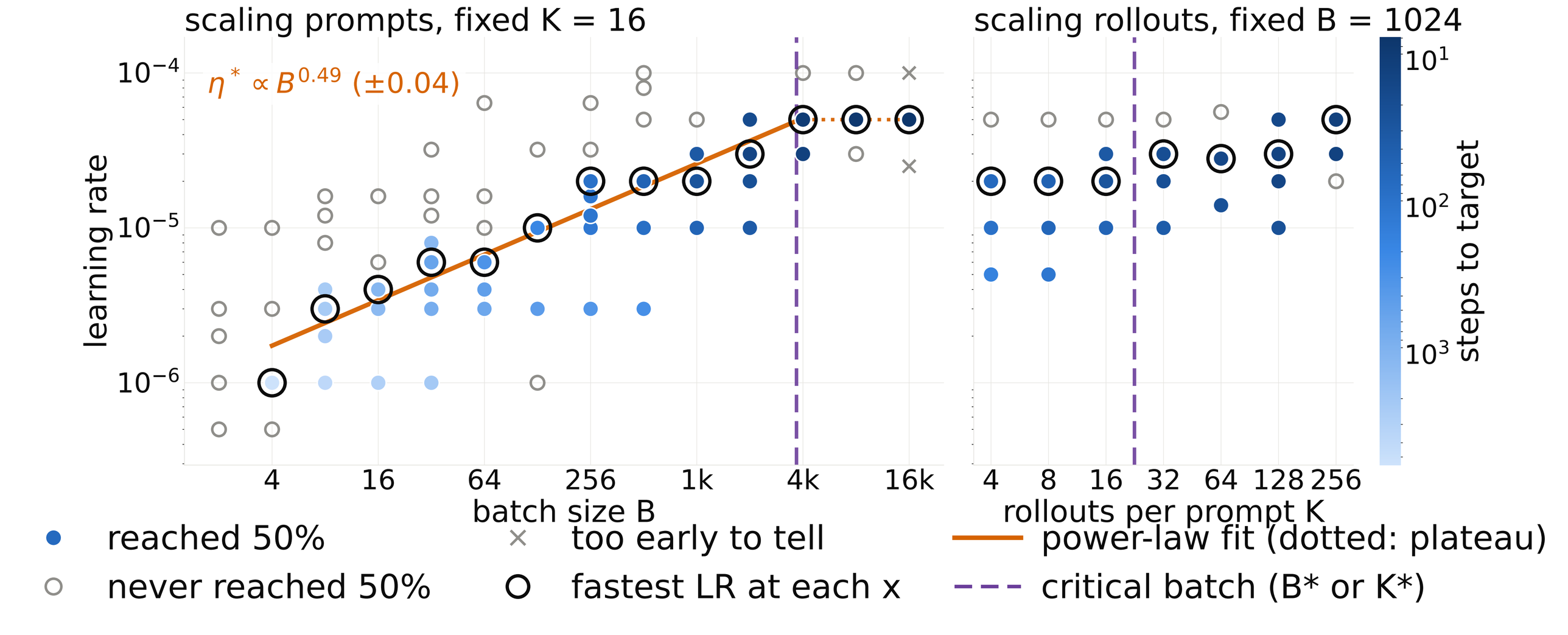
scaling prompts, fixed K = 16
scaling rollouts, fixed B = 1024
$\eta^* \propto B^{0.49}$ (±0.04)
learning rate
10^{-4}
10^{-5}
10^{-6}
4
16
64
256
1k
4k
16k
batch size B
4
8
16
32
64
128
256
rollouts per prompt K
10^1
10^2
10^3
steps to target
reached 50%
never reached 50%
too early to tell
fastest LR at each x
power-law fit (dotted: plateau)
critical batch (B* or K*)